

ABHISHEK KUMAR

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Education

The University of Manchester	United Kingdom
Masters of Science in Data Science - Distinction (Achieved)	09/2023 - 09/2024
• Relevant Coursework: Statistics and Machine Learning, Databases, Understanding Data & its Environment, Financial Data Analytics, Text Mining.	
SRM Institute of Science and Technology	India
Bachelors of Technology in Information Technology - Distinction (Achieved)	05/2016 - 05/2020
• Relevant Coursework: Python, R Programming, Data Science and Big Data Analytics, DBMS, Data Mining, Data Warehousing, Statistics.	

Experience

Data Scientist - Finance MQube	03/2024 - 09/2024
• Developed production-grade ML pipelines for financial market prediction using AWS SageMaker and Lambda, aligned with MLOps best practices. • Fine-tuned domain-specific LLMs for sentiment analysis and returns forecasting, enhancing risk-adjusted returns by 30%. • Used quantitative modeling and rolling statistical windows to develop adaptive ML pipelines responsive to real-time financial trends. • Engaged stakeholders through model visualisations and performance dashboards to support operational decision-making.	
Data Scientist - Geospatial Equans	11/2023 - 02/2024
• Analyzed Scope 3 emissions across multi-modal travel data (air, train, car, accommodation) using Databricks and MySQL for data exploration and ETL, with ArcGIS and Power BI for geospatial analysis and dashboarding, optimizing routes to reduce CO ₂ emissions by 75% and costs by 20%. • Developed automated ETL pipelines and applied geospatial analysis (Shapely, GeoPandas, ArcGIS, NetworkX) to map high-emission routes, align corporate travel policies with sustainability goals, and drive data-driven decision-making for business decision-makers.	
Data Scientist - Fraud and Risk National Payments Corporation of India	11/2020 - 08/2023
• Developed 30+ fraud prediction models using Random Forest, Autoencoders, and XGBoost, reducing fraud by 35% and false positives by 20%. • Built end-to-end ML pipelines on Azure, using Azure Data Factory for ETL, Azure ML for model training/versioning, and AKS for deployments. • Engineered scalable data ingestion workflows using Kafka and Azure Event Hubs to support processing of up to 2 billion UPI transactions monthly. • Built a churn prediction model using user activity patterns, transaction frequency, and engagement drop-offs to forecast user retention risk. • Performed SQL-based feature extraction, trend analysis, and fraud signal engineering on multi-million-row financial datasets. • Applied statistical hypothesis testing, probability distribution analysis, and sampling methods to evaluate fraud model performance and validate precision-recall trade-offs. • Led A/B testing and, designed KPI-driven Power BI dashboards using Power Query to help decision-makers interpret model performance. • Maintained production codebases and translated complex ML outputs into actionable insights for technical and non-technical stakeholders.	

Projects

Airbnb Listing Search Agent with Google Gemini and MultiMCP
• Built an AI agent to automate Airbnb listing and geolocation searches, streamlining travel planning across multiple platforms. • Integrated Google Gemini 2.5 with MultiMCPTools (Airbnb + Google Maps servers) using async Python, enabling real-time, tool-augmented responses to natural language queries.
DocuChat- RAG
• Built an Agentic RAG chatbot using CrewAI, LangChain, Qdrant, and Streamlit for Q&A on PDFs and web search. • Integrated local LLM inference (Ollama), prompt engineering, and fallback search (FireCrawl) for dynamic knowledge retrieval.
TextScout: Customer Transcript Analysis Using NLP and Generative AI
• Built an NLP pipeline using RoBERTa, BART (HuggingFace), FAISS, and prompt engineering for real-time sentiment and zero-shot classification. • Implemented self-learning AI with automated retraining triggered by FAISS, improving model classification accuracy.
FilmateAI – AI-Powered Movie Recommendation System
• Built a movie recommendation engine using OpenAI embeddings, Groq's Mixtral LLM, and Qdrant for semantic search over movie metadata. • Developed a full-stack app with Flask (backend) and React + TypeScript (frontend), integrating GPT-3.5 for sentiment filtering and enabling personalized, tone-aware movie recommendations.
Fraud Detection in Credit Card Transactions
• Built an unsupervised anomaly detection pipeline using Python, NumPy, Scikit-Learn, and unsupervised learning methods for fraud detection. • Engineered robust feature scaling and PCA for dimensionality reduction, achieving 84% recall while minimizing false positives.

Skills

Technical Skills:
Python (Pandas • NumPy • Matplotlib • Seaborn • Plotly • PyTorch • scikit-learn) • TensorFlow • R • Java • PowerBI • DAX • Advanced SQL • PostgreSQL • MongoDB • ETL • Machine Learning • Regression • Clustering • Time Series Forecasting • Predictive modeling • Anomaly Detection • Deep Learning • Neural Networks • Natural Language Processing (NLP) • LLMs • RAG • Prompt Engineering • LangChain • AgenticAI • MCP • AWS (Sagemaker • S3 • Lambda • Quicksight) • Azure • Databricks • PySpark • Version Control (Git) • Linux • Docker • MLOps • MLFlow • CI/CD • FastAPI/Flask.